
Lab Environment Setting Practice

Join GitHub classroom & Install riscv-xv6

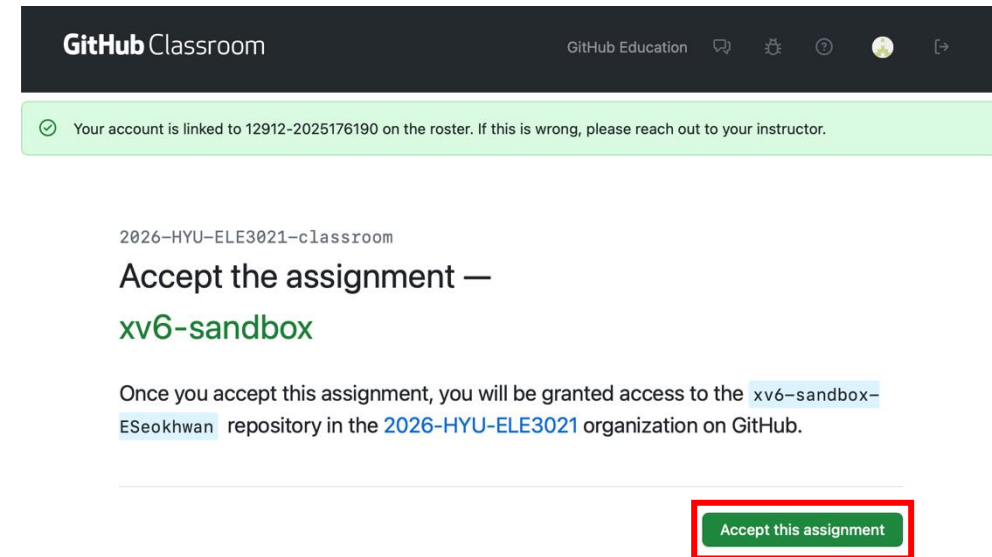
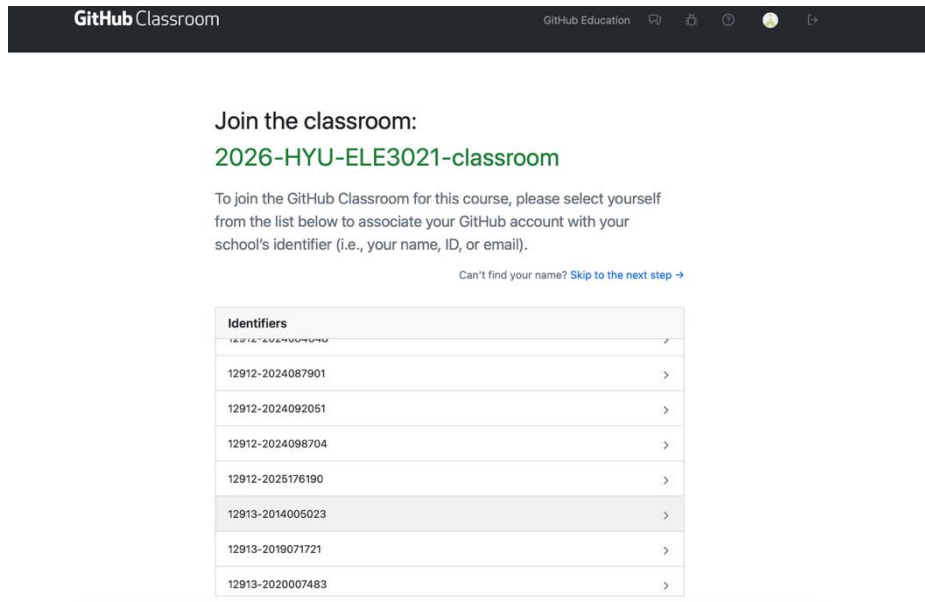
Originated by Database Operating Systems Lab.,
Revised by Software Platforms Lab.

Join GitHub Classroom

- Access the following link to create a repository for the labs:
 - <https://classroom.github.com/a/mexPtowY>
 - We will continue to use this repository for the in-class hands-on labs.
 - For the projects, a new link will be provided for each.
- If you do not have a GitHub account, please sign up for one.

Join GitHub Classroom

- Select your identifiers from the list and accept the assignment.
- Identifier Format: [class_code]-[student_id]



- If you can not find your identifier or made an incorrect selection, please contact the TA.

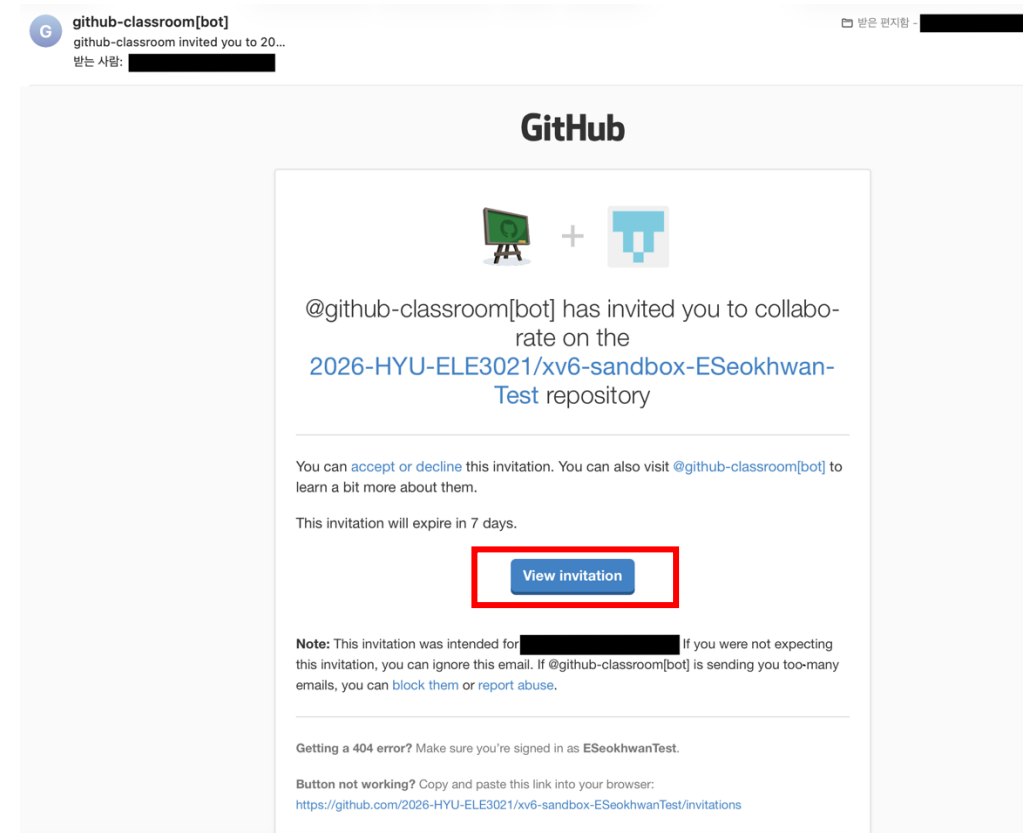
Join GitHub Classroom

- If you encounter an access error (left), check your e-mail and accept the invitation (right).

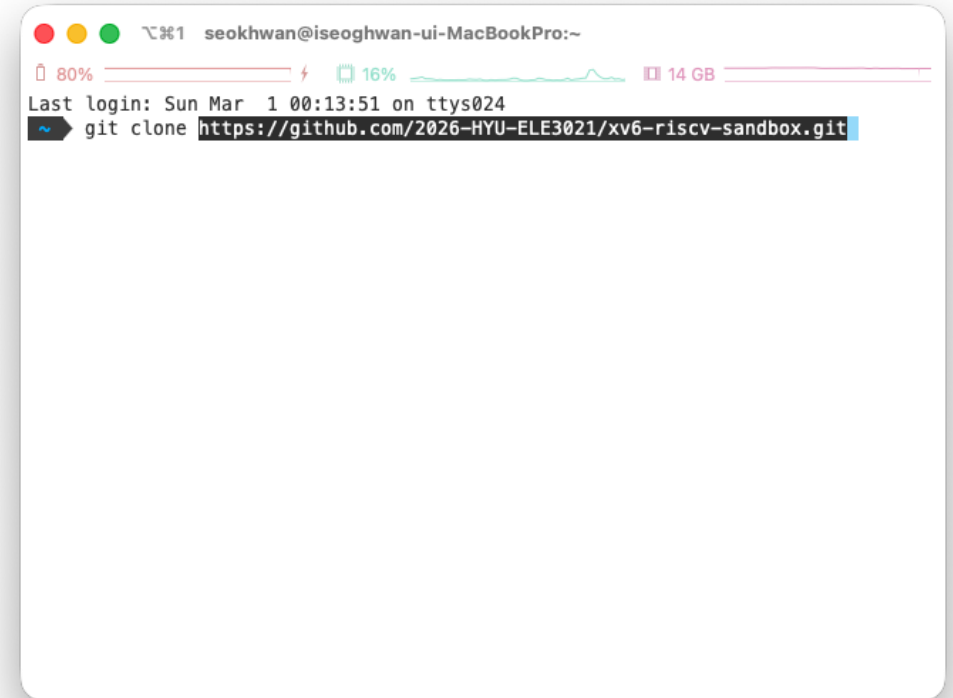
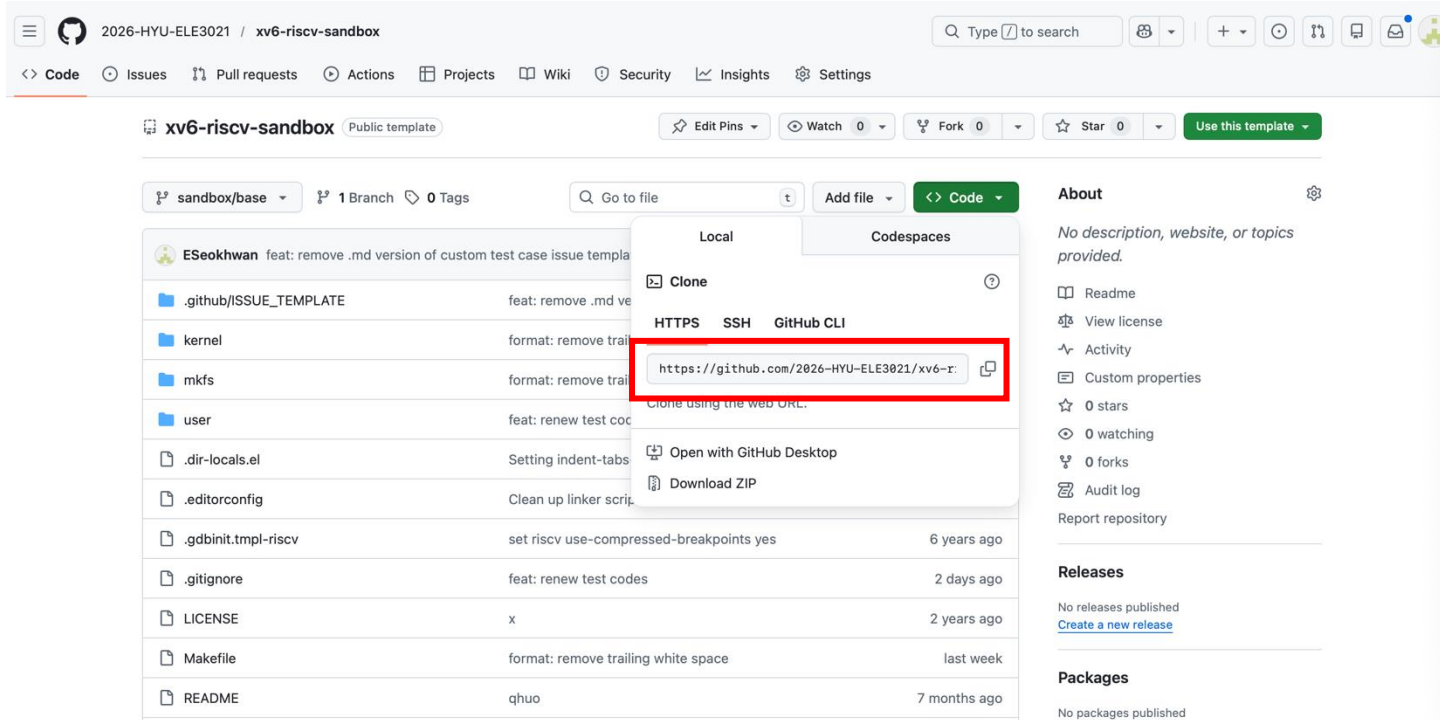


Repository Access Issue

You no longer have access to your assignment repository. Contact your teacher for support



Clone your repository



- Copy the link of the repository and clone it using your **Linux/Mac** terminal.

❖ See **Appendix** for: **Linux environment setup** on windows laptop, **Git Installation, Git authentication setup**

Install RISC-V Compiler

- RISC-V (Reduced Instruction Set Computer V)
 - A **simple and modern CPU instruction dictionary** (ISA) that anyone can use for free.
 - It avoids the complexity of legacy architectures like x86.
- Install RISC-V Compiler toolchains

- Linux

```
$ sudo apt update
$ sudo apt install -y build-essential gdb-multiarch gcc-riscv64-linux-gnu binutils-riscv64-linux-gnu
$ riscv64-linux-gnu-gcc --version # install verification
```

- MacOS

```
$ brew update
$ brew tap riscv/riscv
$ brew install riscv-tools
$ riscv64-unknown-elf-gcc --version # install verification
```

❖ See **Appendix** for: Install brew

Install QEMU

- QEMU (Quick EMUlator)
 - A **software emulator** that simulates a RISC-V CPU on your local machine (x86 or ARM).
 - It allows us to run and debug the RISC-V xv6 operating system without needing actual RISC-V hardware.

- Install RISC-V Compiler toolchains

- Linux

```
$ sudo apt update
$ sudo apt install -y qemu-system-misc
$ qemu-system-riscv64 --version # install verification
```

- MacOS

```
$ brew update
$ brew install qemu
$ qemu-system-riscv64 --version # install verification
```

❖ See **Appendix** for: Install brew

Build & Run xv6

- **Navigate to the root directory of your cloned repository**

```
$ cd <your-repository-path>
```

- **Build and run xv6 on QEMU**

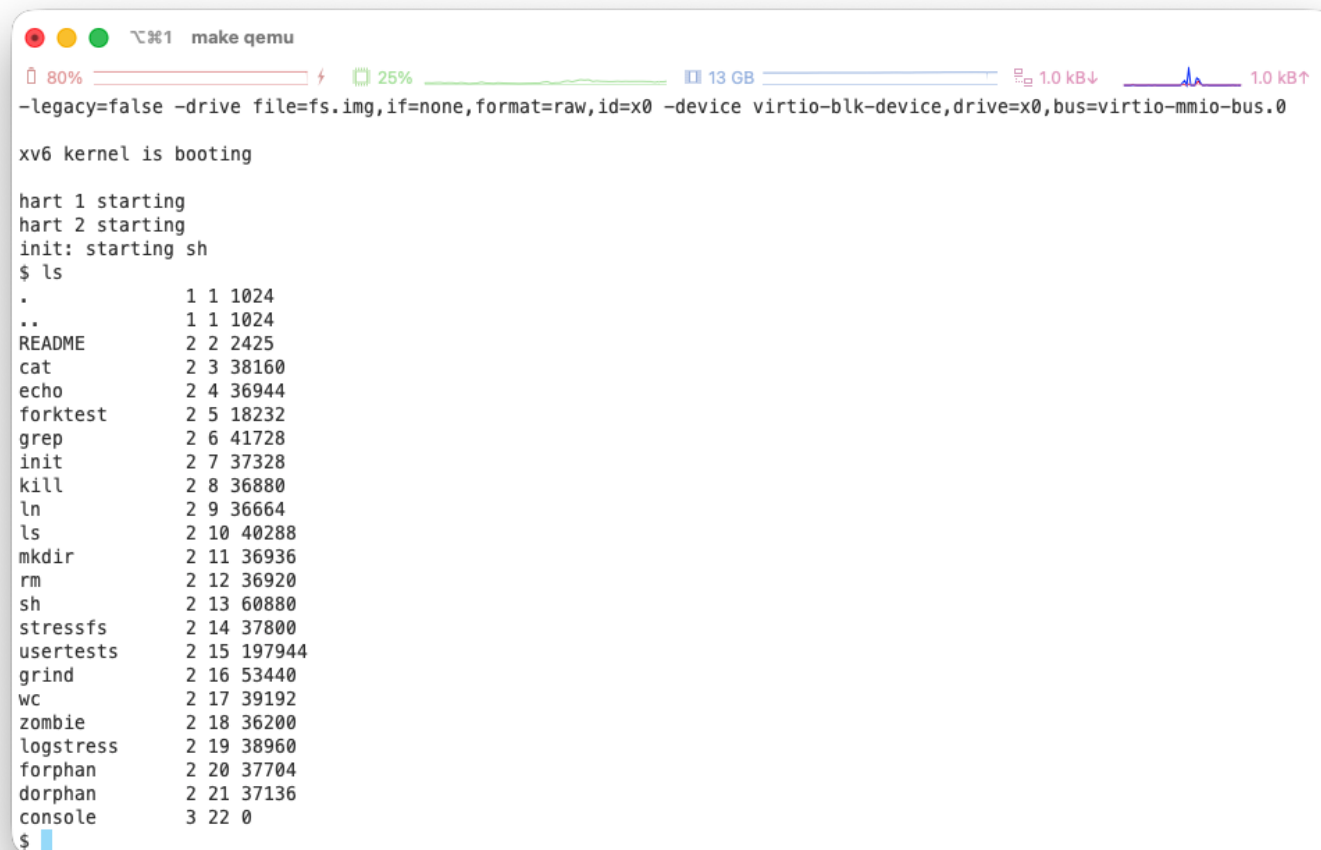
```
$ make qemu
```

- **How to exit QEMU**

```
Ctrl-A, X
```


Today's In-Class Hands-on Lab

- Run xv6 on QEMU and check out what's in the root directory of xv6 operating system using *ls* command.



```
make qemu
80% 25% 13 GB 1.0 kB↓ 1.0 kB↑
-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

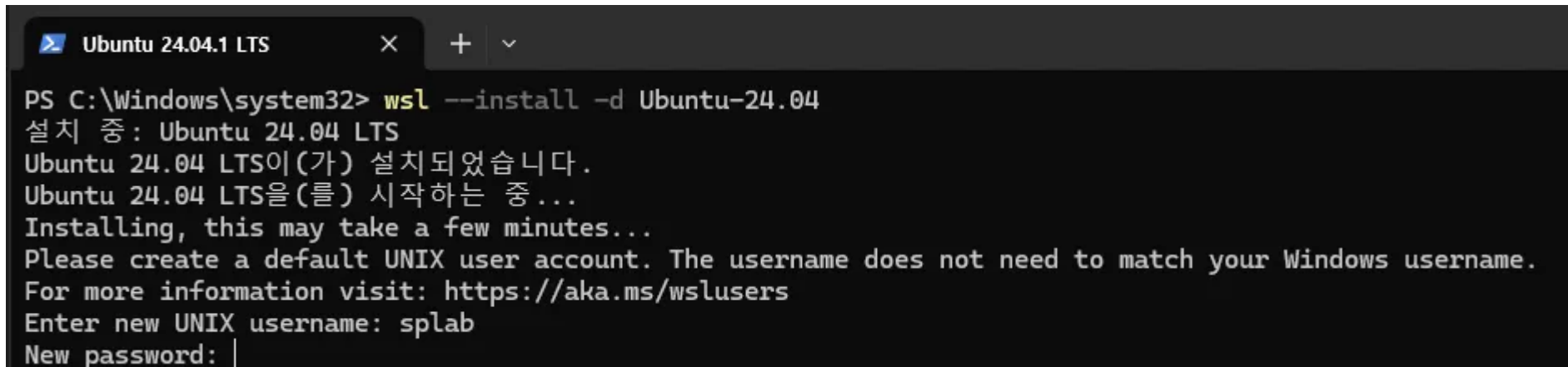
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ ls
.          1 1 1024
..         1 1 1024
README    2 2 2425
cat        2 3 38160
echo      2 4 36944
forktest  2 5 18232
grep       2 6 41728
init       2 7 37328
kill       2 8 36880
ln         2 9 36664
ls         2 10 40288
mkdir     2 11 36936
rm        2 12 36920
sh        2 13 60880
stressfs  2 14 37800
usertests 2 15 197944
grind     2 16 53440
wc        2 17 39192
zombie    2 18 36200
logstress 2 19 38960
forphan   2 20 37704
dorphan   2 21 37136
console   3 22 0
$
```

Appendix

Install Ubuntu on WSL2 (For Windows users)

- We will use **WSL**(Windows Subsystem for Linux), compatibility layer for running Linux binary executables natively on Windows
- Use powershell to install
 - You must install version **24.04** to run xv6-riscv

```
$ wsl --install -d Ubuntu-24.04
```



```
Ubuntu 24.04.1 LTS
PS C:\Windows\system32> wsl --install -d Ubuntu-24.04
설치 중: Ubuntu 24.04 LTS
Ubuntu 24.04 LTS이(가) 설치되었습니다.
Ubuntu 24.04 LTS을(를) 시작하는 중...
Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: splab
New password: |
```

Install GIT

- Linux
 - Open your terminal and run the following commands:

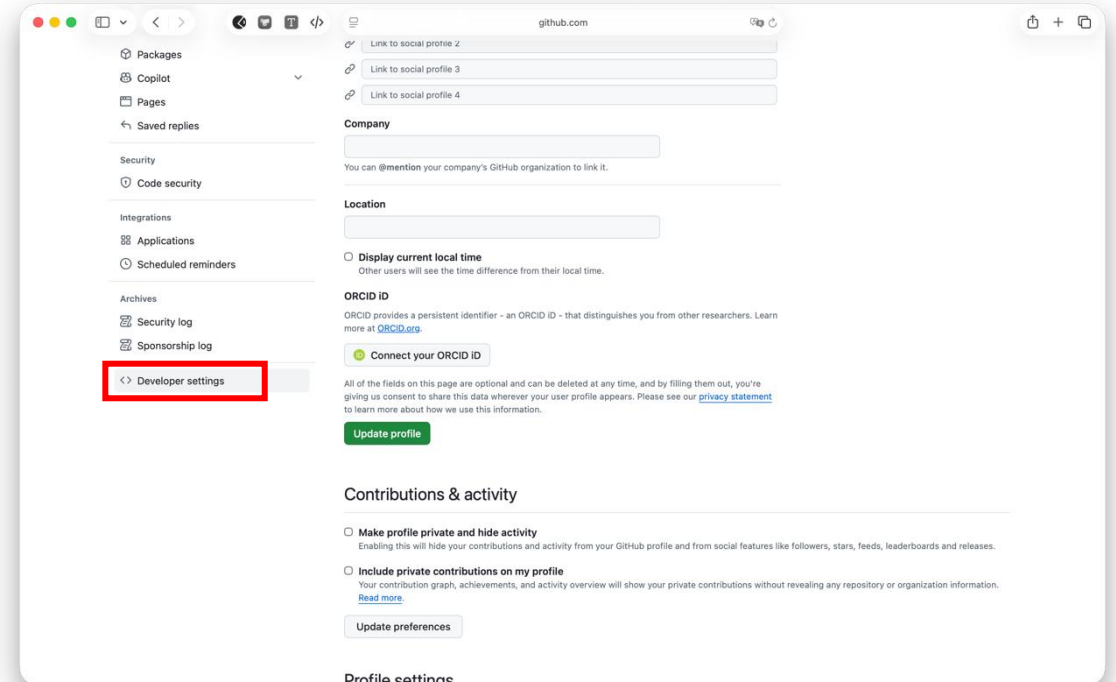
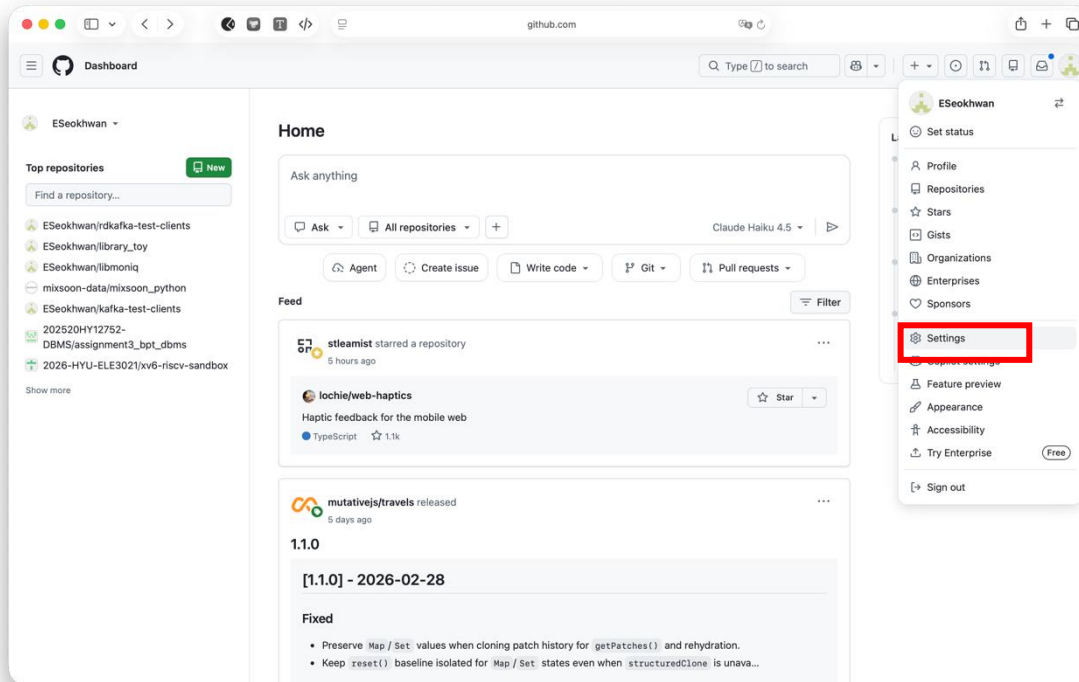
```
$ sudo apt update  
$ sudo apt install git  
$ git --version # install verification
```

- MacOS
 - Open the Terminal app and run the following commands. If Git is not installed, a pop-up will appear prompting you to install it.

```
$ git --version
```

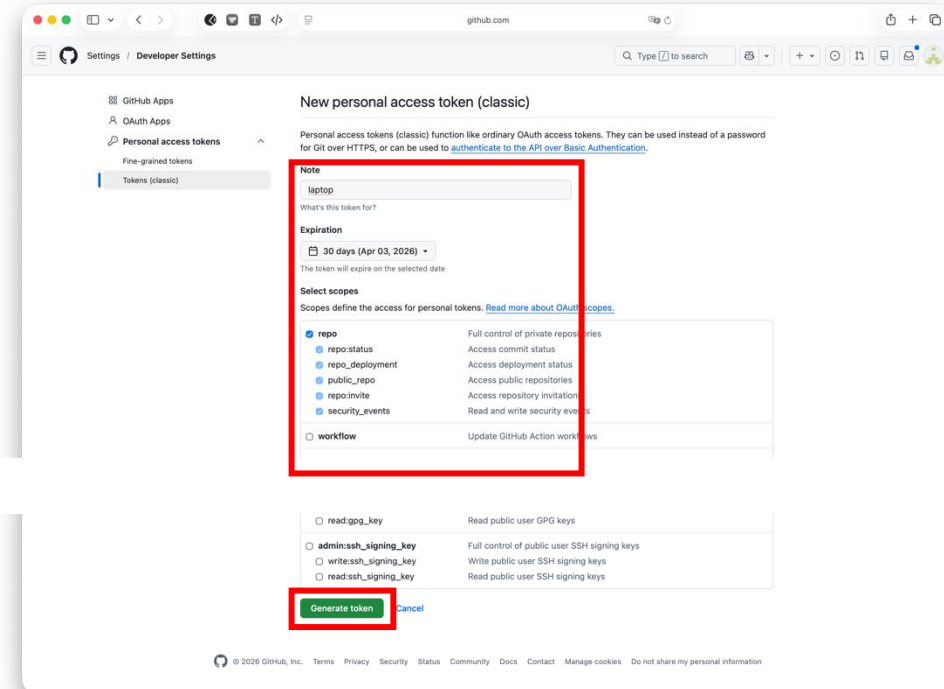
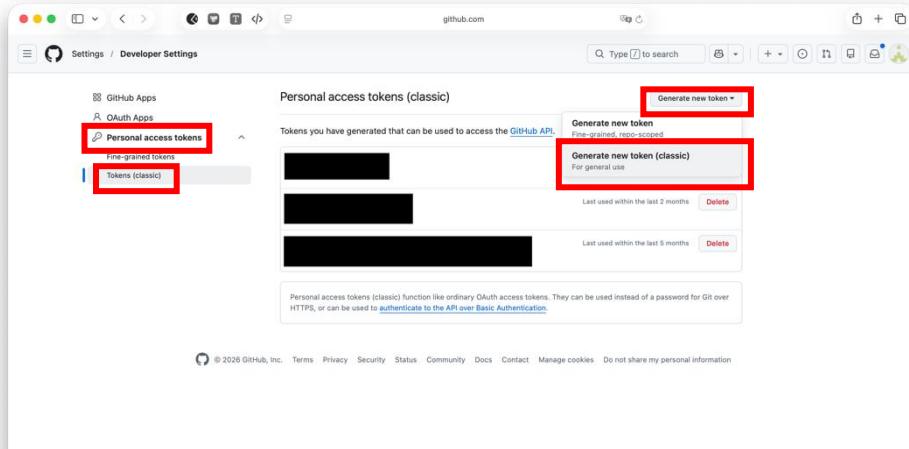
Initial Git Authentication Setup

- Navigate to Developer Settings

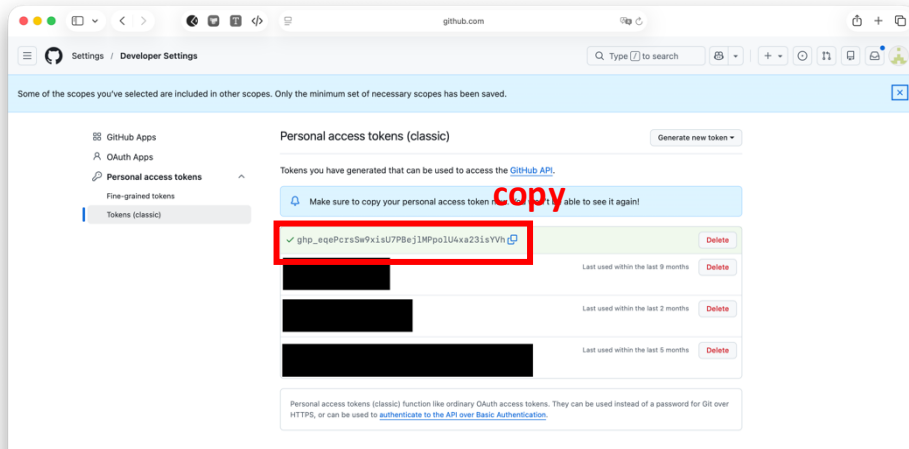


Initial Git Authentication Setup

- Generate token



- Copy token and paste it on password



Install Homebrew (For MacOS users)

- **Homebrew**

- The essential package manager for MacOS.
- It allows you to install developer tools like riscv-tools and qemu easily.

- **Install Homebrew**

- Paste the following command and press Enter

```
$ /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

- Follow the on-screen instructions
- Verify install like below

```
$ brew --version # install verification
```